

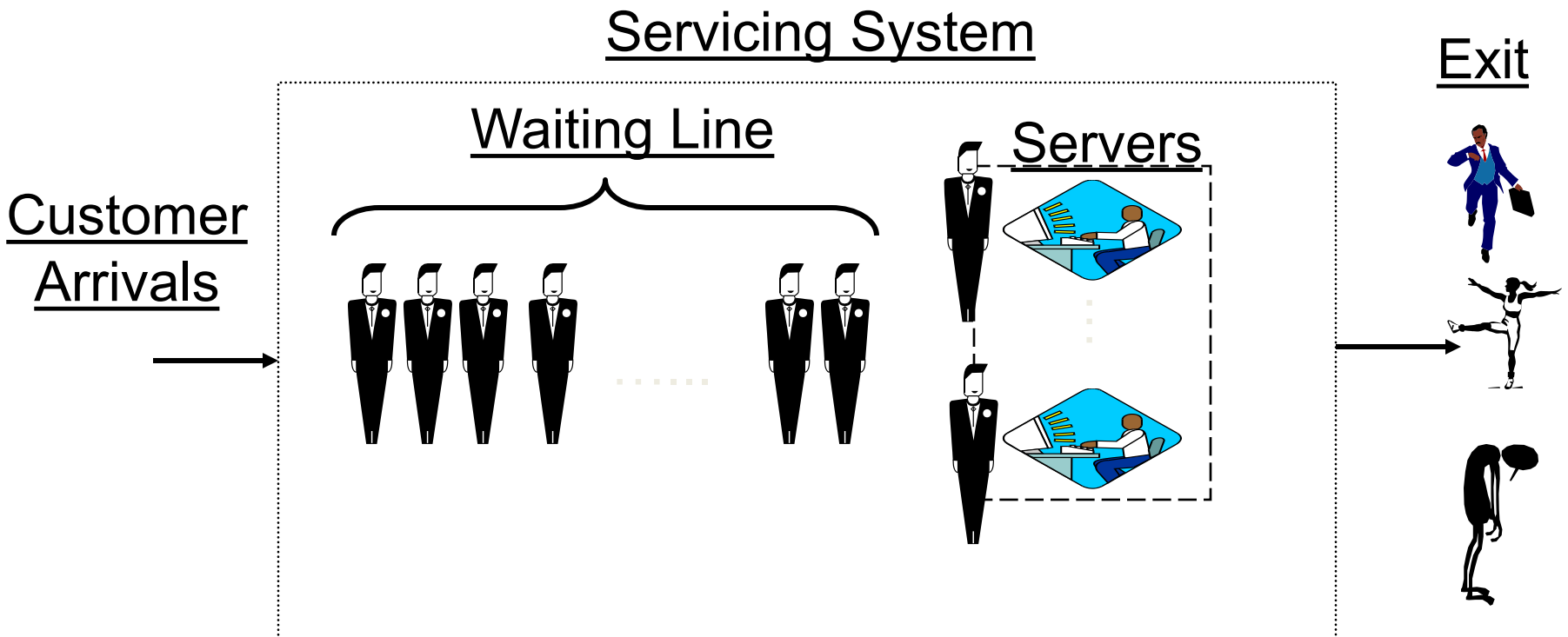
Nanyang Business School
BU5401:
Management Decision Tools

Seminar 8:
Waiting Line Models I

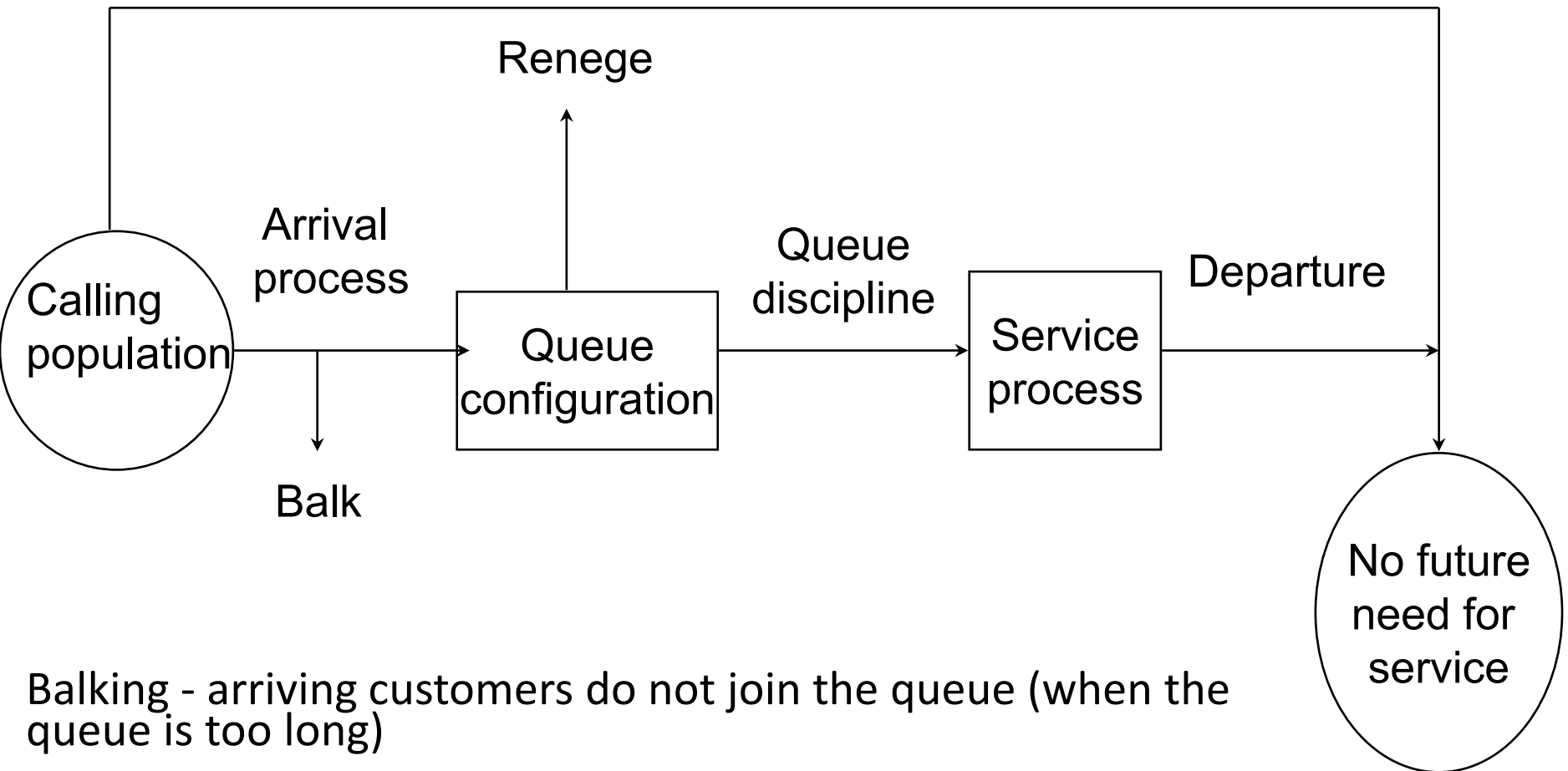
Outline

- Structure of Waiting Line System
- Queuing Systems Parameters & Notations
- M/M/1 Waiting Line Model
- M/G/1 Waiting Line Model
- M/D/1 Waiting Line Model

Structure of Queues



Essential Features of a Queue



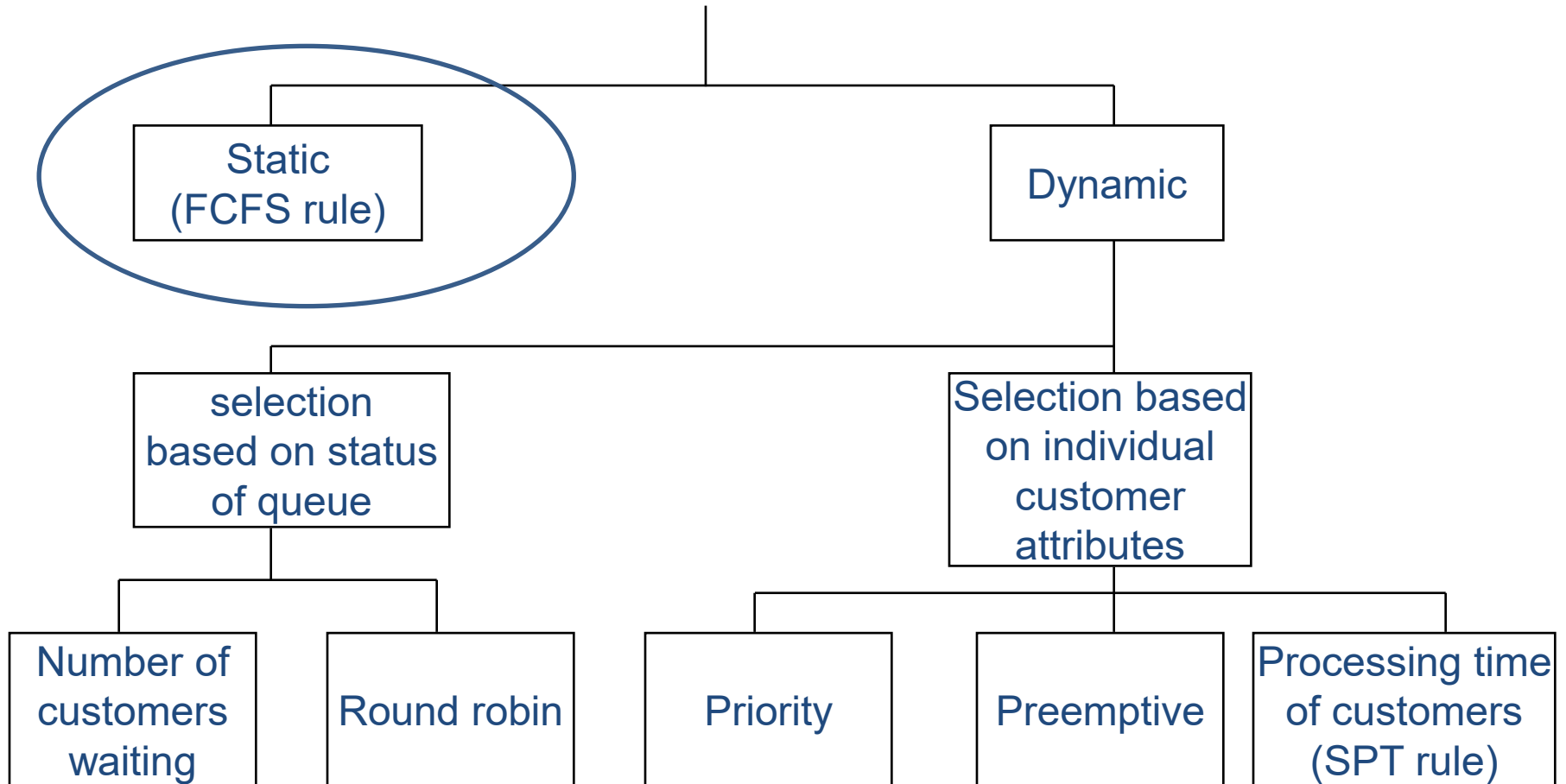
Balking - arriving customers do not join the queue (when the queue is too long)

Reneging - customers give up and leave after waiting for a while

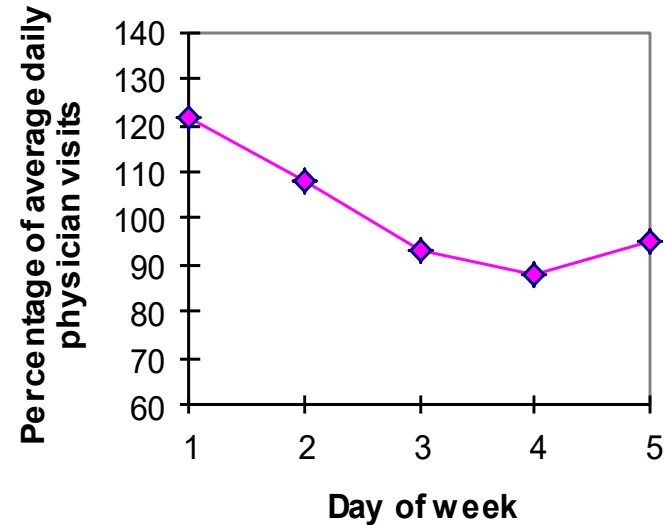
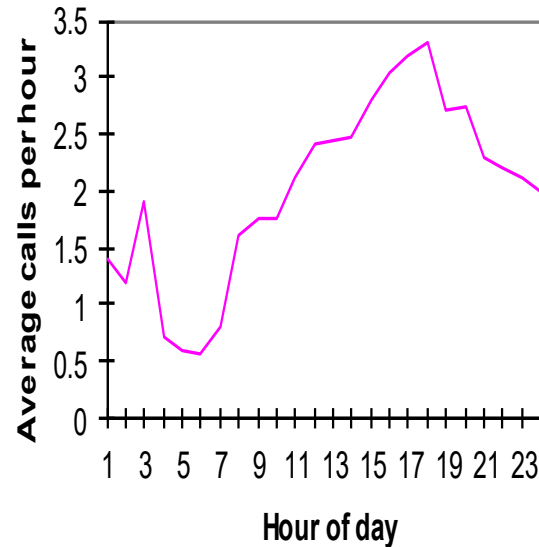
Queues study Steady State Equilibrium (SSE) behaviors

Queuing Discipline

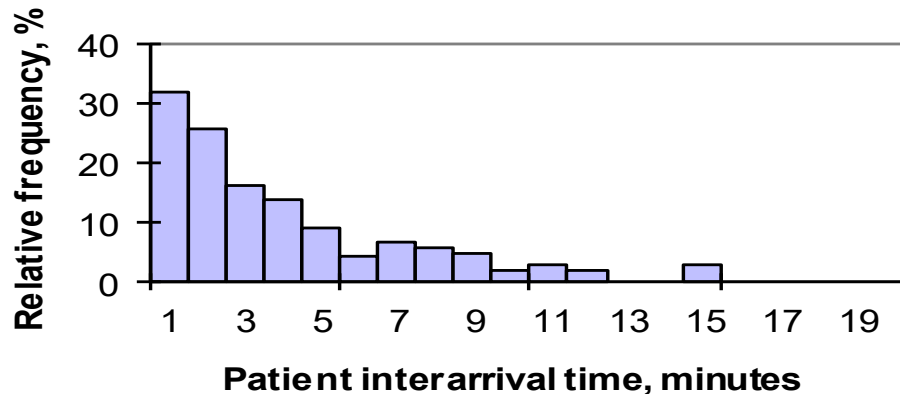
The order in which customers in queue are served



Temporal Variations in Arrival Rate



Distribution of Inter-Arrival Time



Arrival rate : Average number of arrivals per unit time

Service Pattern

- Distribution for the time to serve a customer, e.g. deterministic, exponential, or any other theoretical or empirical distribution
- Service rate = average number of customers served per unit time
- Exponential service time implies Poisson service rate and vice versa

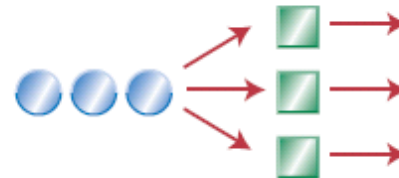
The Service System

Configuration: How the service system is set up

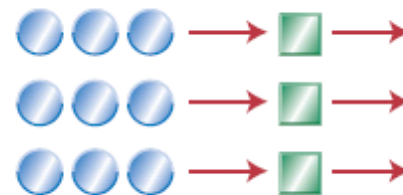
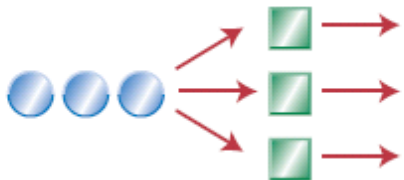
1. Number of phases (single vs. multiple)



2. Number of servers (channels) (single vs. multiple)



3. Number of waiting lines (single vs. multiple)



Kendall-Lee Notation for Queuing Systems

$A/B/k$ is used to describe various queuing systems where
 A identifies the arrival distribution,
 B the service (departure) distribution and
 k the number of channels for the system.

Symbols used for the arrival and service processes are:

M - Markov distributions (Poisson/exponential),

D - Deterministic (constant) and

G - General distribution (with a known mean and variance).

Performance of Queuing Systems

L_q : Average queue length

W_q : Average time in queue

L : Average number of customers in the system

W : Average time in system

λ : Arrival rate, average number of arrivals per unit time

P_0 : Probability that no one is in the system

μ : Service rate, average number of customers served per unit time

P_n : Probability that there are n customers present in the system

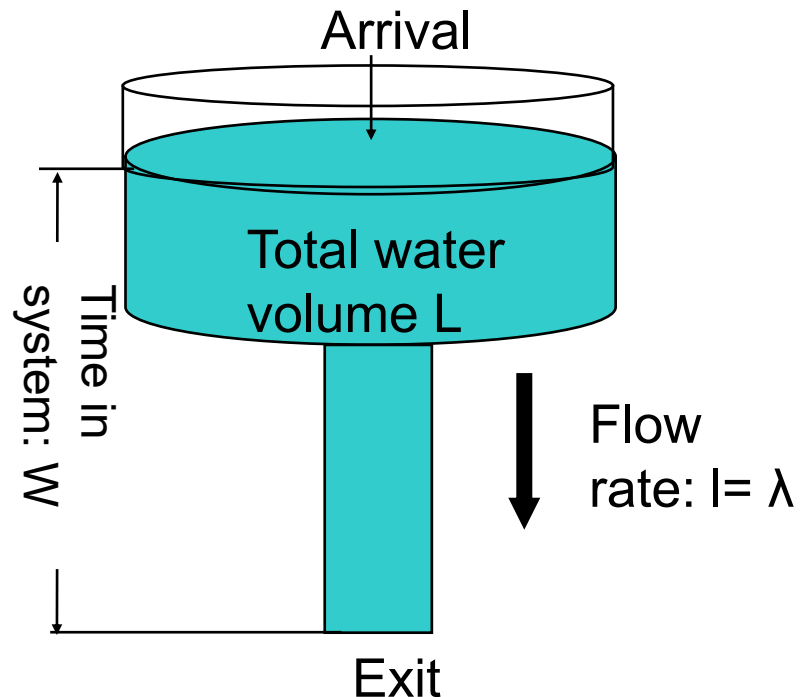
ρ : Service utilisation factor (proportion of time server is busy)

Little's Law

For nearly all queuing systems, there is a relationship between the average time a unit spends in the system or queue and the average number of units in the system or queue.

These relationships, known as Little's flow equations are:

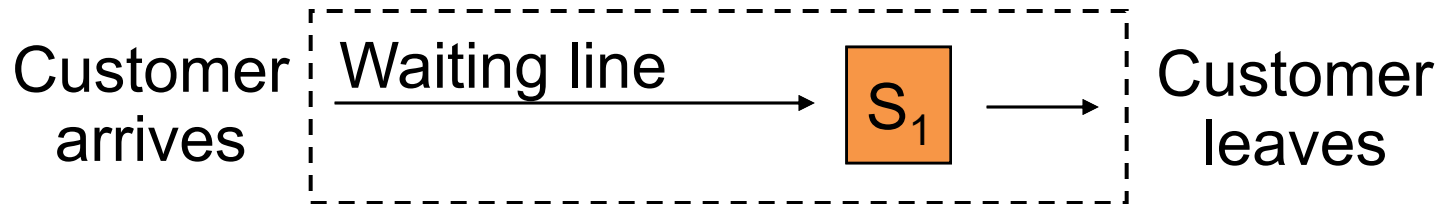
$$L = \lambda W \text{ and } L_q = \lambda W_q$$



Single Server Queues

- The M/M/1 Queue
- The M/G/1 Queue
- The M/D/1 Queue

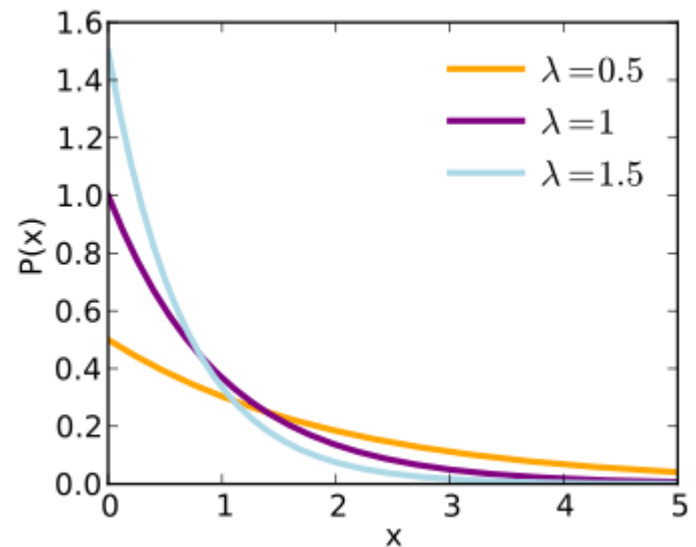
Single Server – M/M/1 Queue



- Infinite calling population
- Infinite queue length
- Poisson Arrivals Rates, Exponential inter-arrival times
- FCFS queue discipline, infinite queue size
- Exponential service time
- Customer are patient (i.e., no reneging)

The Exponential Distribution

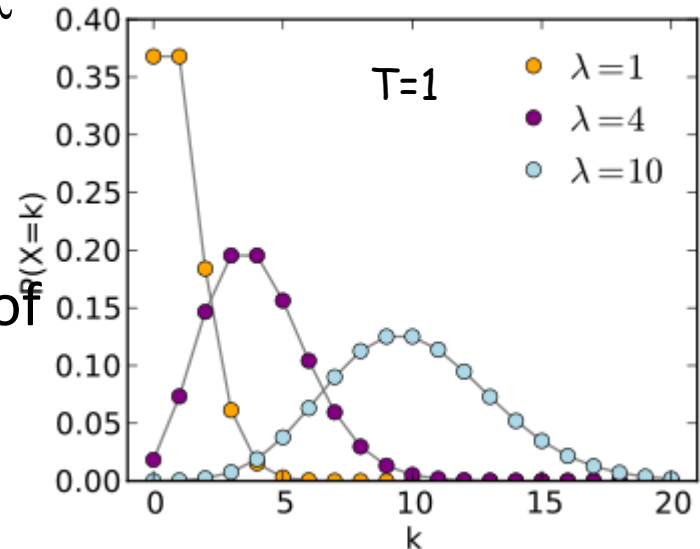
- Arrival rate: λ (on average, there are λ arrivals per unit time)
- Inter-arrival time X : time between two consecutive arrivals
- No memory property: future does not depend on past
- Probability density function:



$$f(t) = \lambda e^{-\lambda t}$$

The Poisson Distribution

- Arrival rate: λ (on average, there are λ arrivals per unit time)
- If inter-arrival time follows an exponential distribution, the number of arrivals (X) in a period of length T follows a Poisson distribution
- Probability distribution function:



$$P(X = n) = \frac{(\lambda T)^n e^{-\lambda T}}{n!}$$

An Example: Joe Ferris

Joe Ferris is a stock trader on the floor of the New York Stock Exchange. Stock transactions arrive at a mean rate of 20 per hour. Each order received by Joe requires an average of two minutes to process.

What is the probability that (a) no orders, (b) exactly 3 orders, and (c) more than 6 orders are received within a 15-minute period?

- Arrival rate for a 15-minute period: $\lambda T = 5$
 - a. $P(x = 0) = (5^0 e^{-5})/0! = e^{-5} = .0067$
 - b. $P(x = 3) = (5^3 e^{-5})/3! = 125(.0067)/6 = .1396$
 - c. $P(x > 6) = 1 - P(X \leq 6)$
Since $P(x = 0) + \dots + P(x = 6) = 0.762$,
 $P(x > 6) = 1 - 0.762 = 0.238$

Steady State Measurements for M/M/1

Formulae

Under assumption of $\rho = \frac{\lambda}{\mu} < 1$

$$P_0 = 1 - \frac{\lambda}{\mu} = 1 - \rho$$

$$P_n = P_0 \left(\frac{\lambda}{\mu} \right)^n = P_0 (\rho^n)$$

$$L_q = \frac{\lambda^2}{\mu(\mu - \lambda)} = \frac{\rho^2}{1 - \rho}$$

$$L = L_q + \frac{\lambda}{\mu} = \frac{\frac{\lambda}{\mu}}{1 - \frac{\lambda}{\mu}}$$

$$W_q = \frac{L_q}{\lambda} = \frac{\lambda}{\mu(\mu - \lambda)}$$

$$W = \frac{L}{\lambda} = \frac{1}{\mu - \lambda}$$

[Little's Law]

An Example: A Toll Road

A toll road has one attendant at an exit gate. Cars arrive in a Poisson fashion at the rate of 120 cars per hour. It takes the attendant, on the average, 15 seconds to service a car. Service times are exponentially distributed. Obtain the basic queue performance measures for the system. Assume that there is an infinite car calling population and an infinite queue length is possible.

Performance of the Gate

$$\lambda = 2 \text{ cars per minute} \quad \mu = 4 \text{ cars per minute}$$

$$\rho = \lambda/\mu = 2/4 = 0.5 \quad P(0) = 1 - \rho = 1 - 0.5 = 0.5$$

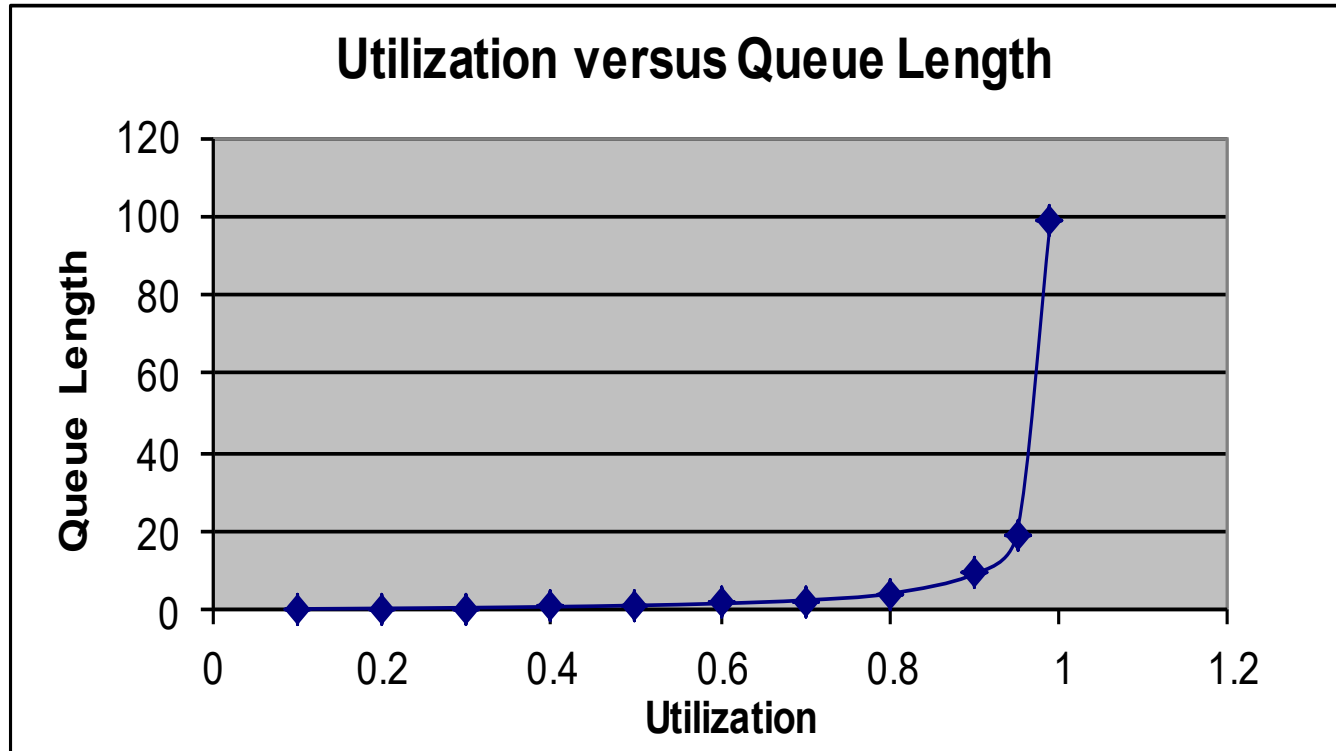
$$L = \lambda/(\mu - \lambda) = 2/(4 - 2) = 1.0$$

$$L_q = \lambda^2/[\mu(\mu - \lambda)] = 2^2/[4(4 - 2)] = 0.5$$

$$W = L/\lambda = 1/2 = 0.5 \text{ min}$$

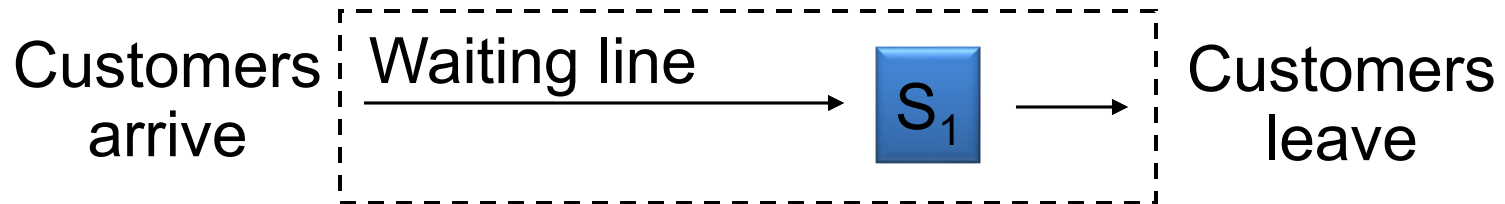
$$W_q = L_q/\lambda = 0.5/2 = 0.25 \text{ min}$$

Impact of Utilization



Note that queue length theoretically goes to infinity when utilization reaches the limit of 1.0.

M/G/1 Queue



*Service time is generally distributed with mean $1/\mu$ and standard deviation σ and under assumption of $\rho = \lambda/\mu < 1$

$$P_0 = 1 - \rho$$

$$P_n = P_0 \rho^n$$

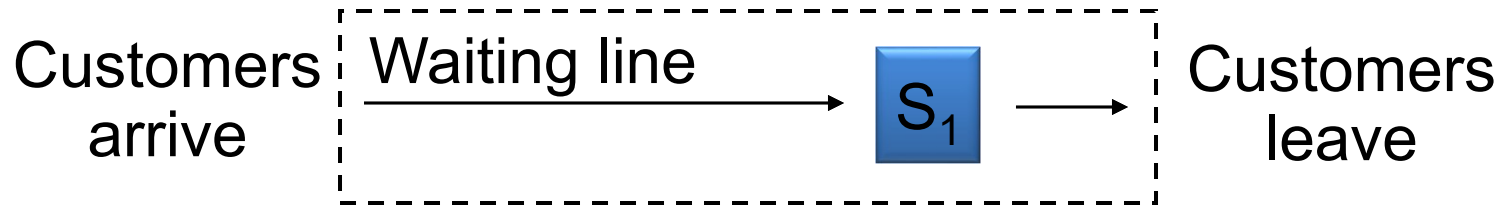
$$L_q = \frac{(\lambda\sigma)^2 + \rho^2}{2(1-\rho)}$$

$$L = L_q + \rho$$

$$W_q = L_q / \lambda$$

$$W = L / \lambda$$

M/D/1 Queue



For Deterministic service time where there is NO variation in service times (ie fixed constant service duration), we set σ to zero in M/G/1 and get:

$$P_0 = 1 - \rho$$

$$P_n = P_0 \rho^n$$

$$L_q = \frac{\rho^2}{2(1-\rho)}$$

$$L = L_q + \rho$$

$$W_q = L_q / \lambda$$

$$W = L / \lambda$$

Note: while the formulas look the same, the numerical values differ from those of M/G/1.